

Multipoint Vibration Localization in Forward Coherent ISAC Enabled by Blind Source Separation

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Abstract We demonstrate multipoint vibration localization in forward ISAC by extracting perturbations directly from the communication signal phase. Leveraging blind source separation, three concurrent vibrations are blindly localized over a 63-km link with meter-level accuracy, alongside penalty-free 30-GBaud 16-QAM transmission. ©2026 The Author(s)

Introduction

The rapid evolution of optical networks toward the 6G era has attracted significant interest in integrated sensing and communication (ISAC). By leveraging deployed fiber-optic infrastructures, ISAC systems enable co-existence of high-speed data transmission and distributed environmental monitoring at minimal marginal cost, such as seismic activity detection and infrastructure health diagnostics [1,2].

In particular, endogenous forward sensing technologies that extract physical perturbation signatures directly from the digital signal processing (DSP) of coherent transceivers have emerged as a high-efficiency solution, offering deep integration without requiring additional sensing hardware or complex modifications to the physical layer [3-6]. Crucially, unlike ISAC schemes based on backscattering distributed acoustic sensing (DAS), forward sensing is inherently compatible with standard erbium-doped fiber amplifiers (EDFAs), overcoming the single-span limitation to enable ultra-long-distance monitoring. Spatial localization in such systems typically relies on a bi-directional architecture, where the position is resolved by the time difference of arrival between signals captured at the East and West receivers. However, existing forward ISAC localization techniques generally struggle to isolate and locate individual vibration sources in multi-event scenarios. Constrained by their reliance on prior spectral knowledge, these approaches lack operational flexibility and easily fail to separate simultaneous, unknown perturbations.

In this paper, we propose and experimentally demonstrate a robust multipoint vibration

extraction and localization scheme in a high-speed coherent communication system based on blind source separation (BSS). By utilizing a degenerate unmixing estimation technique (DUET) framework, we blindly discover and cluster the center frequencies of three independent vibration sources in the time-frequency (TF) domain without any prior knowledge [7]. Experimental results over a 63-km fiber link show that we simultaneously achieve 30-GBaud penalty-free transmission and localization of multipoint perturbations with high spatial resolution and superior stability, offering a promising ISAC solution for next-generation coherent optical networks.

Operation Principle

The principle of the proposed DUET-based forward ISAC scheme is illustrated in Figure. 1. The system adopts a bidirectional communication configuration to precisely locate the multipoint vibration. At the coherent receiver, the demodulated bi-direction phase sequences $\Phi_{\text{East}}(t)$ and $\Phi_{\text{West}}(t)$ are directly extracted from the carrier phase recovery (CPR) stage, which captures real-time perturbations along the fiber link, achieving a high degree of hardware sharing between communication and sensing. Subsequently, these extracted phase sequences are fed into DUET algorithm to resolve the overlapping phase fluctuations in multi-event scenarios.

For a specific physical perturbation, its absolute position z_k is determined by:

$$z_k = (L - v\tau_k) / 2 \quad (1)$$

where τ_k represents the relative time delay between the two counter-propagating signals, L is

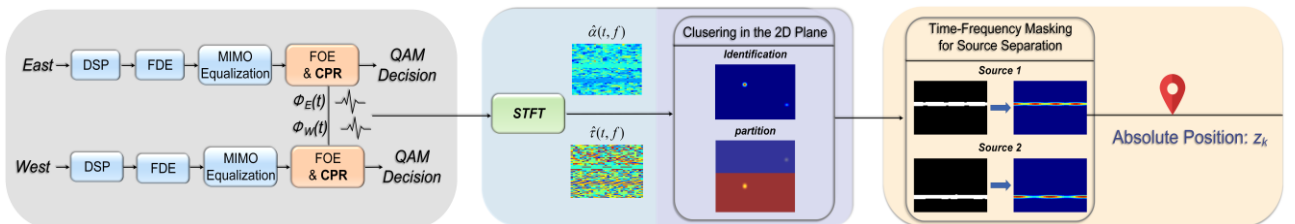


Fig. 1: System architecture and the DUET principle.

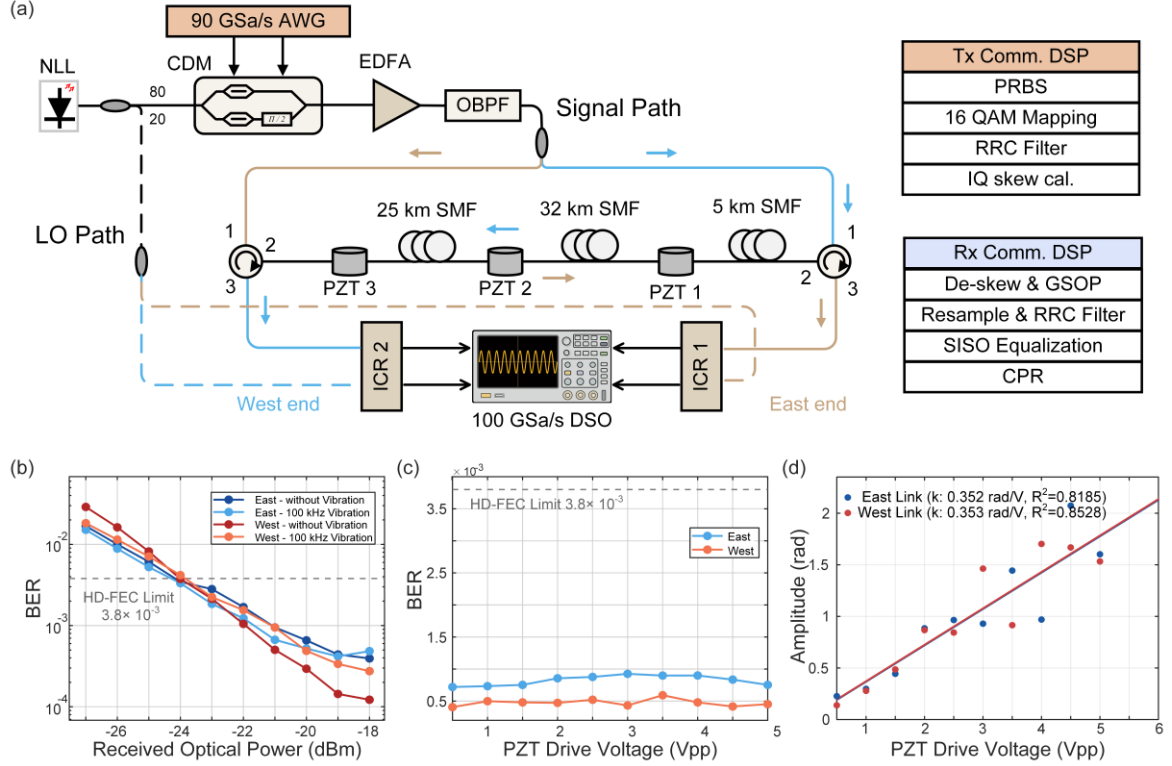


Fig. 1: Experimental setup and communication results.

the total link length and v is the group velocity. The algorithm first employs the short-time Fourier transform (STFT) to map $\Phi_{\text{East}}(t)$ and $\Phi_{\text{West}}(t)$ into the 2D time-frequency (TF) domain, $S_E(t, f)$ and $S_W(t, f)$. Relying on the W-disjoint orthogonality of independent sources, the spatial features are captured by constructing the relative attenuation coefficient matrix $\hat{\alpha}(t, f)$ and the delay matrix $\tau(t, f)$:

$$\hat{\alpha}(t, f) = \ln \left| \frac{S_E(t, f)}{S_W(t, f)} \right| \quad (2)$$

$$\hat{\tau}(t, f) = \frac{\angle[S_W(t, f)S_E^*(t, f)]}{2\pi f} \quad (3)$$

The extracted features (α, τ) from all TF bins are mapped into a 2D parameter plane to construct a power-weighted histogram $H(\alpha, \tau)$. Since TF bins dominated by the same source share consistent spatial fingerprints, they form distinct clusters in the parameter space. By performing peak detection on the histogram, the algorithm blindly discovers the cluster centers (α_k, τ_k) corresponding to independent vibration sources. A binary TF mask $M_k(t, f)$ is then constructed to assign each bin to the vibration source with minimum matching error and enable unmixing of concurrent vibration events. Consequently, the optimal time delay τ_k is identified at each cluster center and the absolute location can be

determined by Eq. (1).

Experimental Setup

The proof-of-concept experimental setup of the proposed ISAC system is illustrated in Fig. 2(a). A narrow-linewidth laser (NLL, NKT E15) is utilized for high-fidelity sensing. The continuous-wave light is subsequently split into two tributaries by an 80:20 optical coupler. The upper tributary is modulated by a coherent driver modulator (CDM, NeoPhotonics Class 60) to generate 30 GBaud 16-QAM single-polarization communication data. The transmission data is then pre-amplified by an EDFA to compensate for insertion loss from the modulator, followed by an optical bandpass filter with 0.8 nm bandwidth to suppress amplified spontaneous emission (ASE). Constrained by the limited number of NLLs and CDMs, the experiment employs co-frequency bi-directional transmission. Thus, the identical modulated signal is equally divided into two paths and launched into the fiber link from both the East and West ends by optical circulators. To emulate multipoint environmental disturbances, three piezoelectric transducers (PZTs) are deployed along the fiber at approximately 5 km, 37 km, and 62 km from the East side, respectively.

The lower tributary serves as the intrinsic local oscillator (LO), which is further split into two branches to feed the coherent receivers at both ends. After propagating through the fiber link, the counter-propagating signals are extracted by the

circulators at the opposite ends. To conduct a single-polarization reception, a polarization controller (PC) is inserted into each receiving path to carefully align the state of polarization, thereby effectively mitigating the polarization fading effect. Two variable optical attenuators (VOAs) are used to adjust the received optical power (ROP) before being detected by integrated coherent receivers (ICRs) at each end. Finally, to facilitate precise clock synchronization between the two bi-directional receivers, one polarization output from each ICR is simultaneously fed into a four-channel digital storage oscilloscope (DSO) and sampled for subsequent offline DSP.

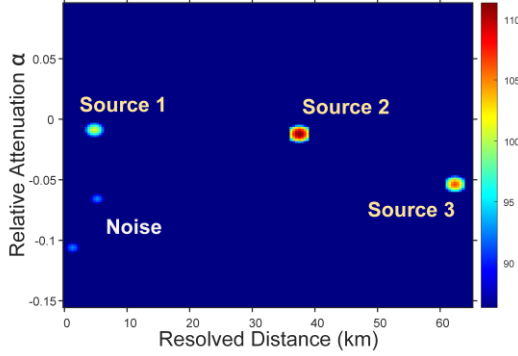


Fig. 3: 2D clustering histogram.

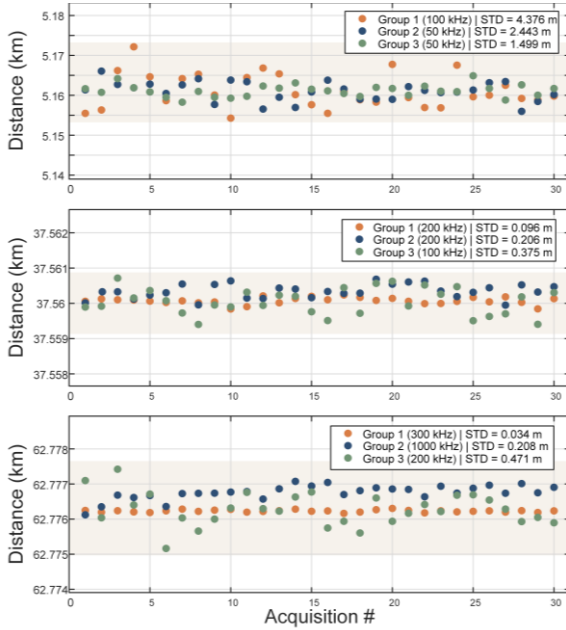


Fig. 4: Three points location results from the east end.

Results and discussions

As shown in Fig. 2(b), we first evaluate the communication integrity of the proposed ISAC system by measuring the bit error rate (BER) performance under different optical received power (ROP). The BER curves for both the East and West links exhibit a tight overlap even when a 100 kHz vibration is applied. Furthermore, Fig. 2(c) demonstrates that varying the driving amplitudes of PZT induces no observable degradation

on the BER. These results confirm that the perturbation has a negligible penalty on the communication performance. Subsequently, the linearity response between the driving amplitude and the phase variation is investigated, as depicted in Fig. 2(d). The extracted phase amplitudes for both links demonstrate an excellent linear proportional relationship with the applied voltage, yielding R^2 values of 0.8185 and 0.8528, respectively. The identical slope responses of the two ends further confirm that the CPR module can reliably restore external perturbations.

Multiple sets of vibration frequencies are strategically adopted across different experimental groups in subsequent multipoint separation and location, which can mitigate the ambiguity and ensure unique localization in the continuous-wave forward sensing schemes. Processed by the DUET-based BSS algorithm, the TF bins are mapped into a 2D parameter space, as shown in the Fig. 3. Three prominent peaks corresponding to Source 1, Source 2, and Source 3 emerge clearly, verifying the capability of the algorithm to blindly discover multiple concurrent events. Finally, the localization stability and uncertainty are assessed by repeating the measurements for 30 acquisitions. As shown in Fig. 4, the resolved locations for all three sources remain highly consistent. For the far-end vibration sources at 32.0 km and 62.7 km, the system consistently achieves sub-meter localization accuracy, with the optimal STD reaching as remarkably low as 0.034 m. For the near-end source around 5.1 km (Source 1), the localization STD ranges between 1.499 m and 4.376 m. This increase in STD is resulted from its intrinsically lower localized sensing SNR, which perfectly aligns with the weaker clustered energy observed in the 2D histogram.

Conclusions

We introduce a DUET-based forward ISAC scheme for multipoint separation and localization requiring no prior knowledge. High-fidelity vibration waveforms are extracted via standard communication DSP along with a negligible penalty to the 30-GBaud 16-QAM transmission. Three independent vibration sources are clearly identified in the 2D clustering histogram, achieving a spatial resolution ranging from 4.37 m down to an impressive 0.034 m at high-SNR positions, providing a promising solution to the long-standing challenge of multipoint localization in forward sensing-based ISAC.

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